

Carlos De La Cruz

Senior Data Scientist | Applied AI, Agentic Systems & ML Products

[Email](#) · [LinkedIn](#) · [GitHub](#) · [Technical writing](#)

[Download PDF resume](#)

Profile

Senior Data Scientist and AI technical lead with 8+ years of experience taking machine-learning and applied-AI initiatives from problem framing and model selection through production delivery. Recent work includes production multi-agent systems, RAG applications for sensitive enterprise knowledge, LLM evaluation, forecasting, recommendation systems and dynamic pricing. Combines hands-on Python and cloud engineering with technical leadership, mentoring and clear communication with business and engineering stakeholders.

Core expertise

- **Applied AI and GenAI:** LLM applications, agent orchestration, RAG, prompt engineering, LLM evaluation, NLP and semantic search
- **Machine learning:** forecasting, recommendation systems, ranking, classification, deep learning, experimentation and Bayesian methods
- **Engineering and MLOps:** Python, SQL, FastAPI, Flask, Docker, CI/CD, MLflow, PostgreSQL/PGVector and Git
- **Cloud and data:** Azure OpenAI, Azure, Databricks, Snowflake, GCP and AWS
- **Frameworks:** AutoGen, LangGraph, LangChain, Hugging Face, PyTorch, TensorFlow and scikit-learn
- **Leadership:** technical direction, architecture decisions, mentoring, code review, stakeholder communication and production handover

Experience

Lead Data Scientist — AI Initiatives | Stora Enso

January 2025 - Present

- Provide technical leadership across AI initiatives, guiding teams from problem definition and model selection through validation and production delivery.
- Define solution architectures and engineering practices for scalable, maintainable and responsible AI systems.
- Collaborate with data scientists, software engineers, engineering leaders and business stakeholders to turn ambiguous needs into

practical AI solutions.

- Mentor data scientists and engineers through architectural guidance, code reviews and production-oriented development practices.
- Evaluate frameworks, models and vendors and translate technical trade-offs into recommendations for technical and non-technical stakeholders.

Senior Data Scientist | Stora Enso

June 2022 - 2025

Production multi-agent decision-support system

- Designed and deployed a production multi-agent AI system supporting commercial negotiations at a global enterprise.
- Built a custom orchestrator extending Microsoft's Magentic-One pattern and coordinating 5+ specialised agents with domain tools and data sources.
- Implemented RAG with PostgreSQL/PGVector, a Snowflake-to-PostgreSQL data pipeline and verification mechanisms for claims generated by agents.
- Developed an evaluation approach using LLM judges, RAGAS and reinforcement-learning-based signals.
- The system has been used by around 10 stakeholders, including decision-makers, and its outputs have supported formal negotiations.

Stack: Python, FastAPI, AutoGen, PostgreSQL/PGVector, Snowflake, Azure OpenAI, Azure, Docker

Production RAG applications for enterprise knowledge

- Designed and delivered production RAG solutions supporting internal teams, including a protected use case that enabled employees to interact with legal documents.
- Led end-to-end implementation with LangChain, Hugging Face, Azure OpenAI, Flask, Docker, PostgreSQL and Azure.

Self-service retail ML products

- Generated the product ideas and developed forecasting, recommendation and dynamic-pricing models for AI-enabled unattended retail systems.
- Delivered the forecasting capability to production using LSTM and ensemble approaches.
- Built collaborative-filtering and hybrid recommender systems and explored Bayesian methods for pricing decisions.
- Was recognised internally as an inventor in a company patent initiative; the US patent process is no longer active.

Stack: Databricks, MLflow, Azure AutoML, scikit-learn, Python, SQL, Snowflake, MongoDB, Azure

Additional applied-ML work

- Applied LDA, BERT, K-means and hybrid approaches to identify themes in safety-notification text.
- Refactored regression workflows and implemented Azure AutoML

SDK for a precision-forestry initiative.

Data Scientist | TUI Global

March 2021 – June 2022

- Developed customer-lifetime-value models using probabilistic and neural-network approaches; delivered results through Snowflake and Tableau.
- Built collaborative-filtering, probabilistic and XGBoost models for personalised destination and hotel recommendations.
- Developed a hotel-search ranking solution combining booking and clickstream signals, with experiment tracking in MLflow.
- Created reusable Python components for automating machine-learning workflows and outputs.

Data Science & Data Analytics | TUI Nordics

April 2019 – March 2021

- Built attribution, customer-segmentation, recommendation and time-series models for e-commerce and marketing decisions.
- Used Python, SQL and BigQuery to analyse customer and web behaviour and translate findings into stakeholder recommendations.
- Created monitoring and decision-support dashboards with Tableau and Google Data Studio.

Guest Lecturer — Data Analytics and Pandas | Hyper Island

December 2020

- Delivered an online guest lecture for data-analytics students on modern analysis workflows with Python and pandas.

Data Analyst — E-commerce | Off The Wall

June 2017 – April 2019

- Conducted web, marketing and conversion analysis and supported experimentation, SEO and campaign optimisation.

Selected publications and public work

- [Building a Comprehensive RAG Application with Azure OpenAI, Flask and LangChain](#)
- [Building Baseline Models with Design Patterns in ML](#)
- [Hyperdimensional Computing: Taking AI to the Next Level by Emulating the Brain](#)
- [Data Ethics and Responsible AI — podcast](#)
- [Data-science portfolio](#)

Education and selected specialisations

- Advanced Machine Learning Specialisation | Higher School of

Economics, 2020-2022

- Data Analysis and Statistics | Nackademin, 2015-2017
- International Relations | Stockholm University, 2010-2015
- Natural Language Processing Specialisation, 2023